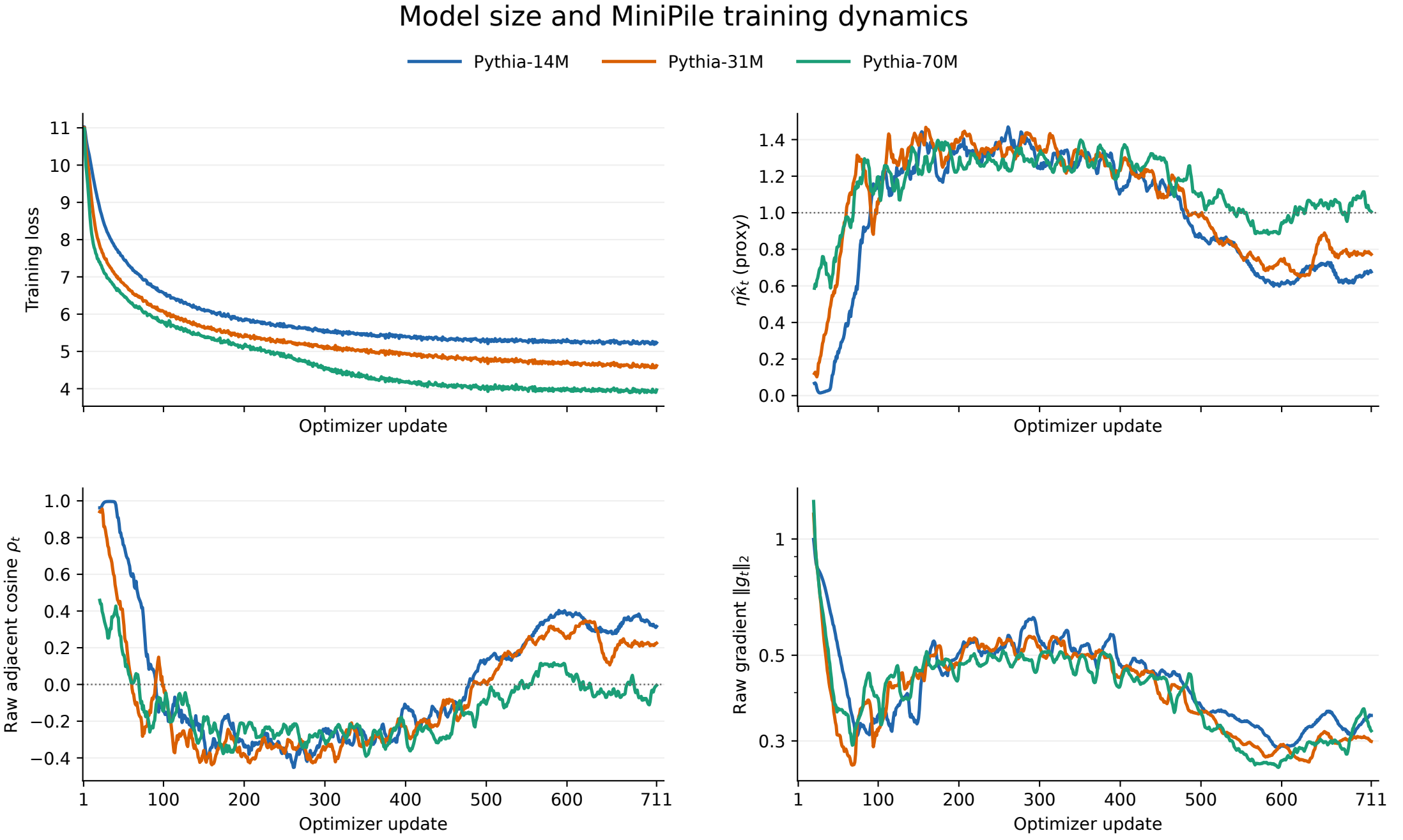

Model size and MiniPile training dynamics
Pythia-14M
Pythia-31M
Pythia-70M
Training loss
Optimizer update
$\hat{\eta\kappa}_t$ (proxy)
Optimizer update
Raw adjacent cosine ρ_t
Optimizer update
Raw gradient $\|g_t\|_2$
Optimizer update
Loss: unsmoothed. Other panels: complete 20-update trailing means. Seed 1235; update 712 excluded.
Loss: updates 1-711. First mean: update 20 (norm), 21 (pairs). Norm: log Y. AdamW q is a proxy.